Roll No.: DL.AI.U4AID25009

Amrita Vishwa Vidyapeetham

School of AI, Faridabad

Mid-Term Examinations, Oct 2025

First Semester

B.Tech AI & Data Science

23MAT106 MATHEMATICS FOR INTELLIGENT SYSTEMS I

Duration: Two Hours | Maximum: 50 Marks

CO	Course Outcomes
CO01	Apply the fundamental concepts of linear algebra and calculus to solve canonical problems analytically and computationally
CO02	Model and simulate simple physical systems using ordinary differential equations
CO03	Apply the concept of probability and random variables to solve elementary problem

Answer all questions

1. **Formulation:**

   1.1 A 2×1 vector is needed to uniquely quantify a direction in 3D space. Determine the size of vector space that will be required to uniquely quantify the following geometrical figures (including their location, orientation and size) in 3D space: (a) a point, (b) a circle, (c) a finite cylinder, (d) a finite right circular cone. [8 marks, CO01, BTL4]

   1.2 We have discussed in the class how to represent audio and visual data as a vector. Determine the vector space to which the following data can be mapped as a single data set: (a) a 10 seconds voice recorder data that collects audio at the rate of 48000 data points per second. (b) a still photo taken by a 1000×1000 pixel camera, (c) a 10 *second* video captured by a 1000×1000 pixel camera taken at 30 frames per second. [6 marks, CO01, BTL4]

2. Consider a system matrix $\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$, whose eigen values are $1, 5, -2, 4$. Determine eigen vectors corresponding to each eigen value. [5 marks, CO01, BTL5]

3. Consider a system which takes in $N \times 1$ vector as input and produces a $N \times 1$ vector as output. If the input vector is denoted as $\vec{A}$ and the output vector as $\vec{B}$, and $\vec{B} = -\vec{A}$. Determine the system matrix for the following cases. [3 marks, CO01, BTL3]

4. Consider a vector $\overrightarrow{a_1} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$. Determine $\overrightarrow{a_2}$ and $\overrightarrow{a_3}$ such that $(\overrightarrow{a_1}, \overrightarrow{a_2}, \overrightarrow{a_3})$ form an orthonormal set. [6 marks, CO01, BTL3]

5. Determine the scalar constants α_1 & α_2 such that the vector $\vec{p}$ can be written as a linear combination of vectors $\overrightarrow{b_1}$ and $\overrightarrow{b_2}$, implying $\vec{p} = \alpha_1\overrightarrow{b_1} + \alpha_2\overrightarrow{b_2}$. [5 marks, CO01, BTL2]

$\vec{p} = \begin{bmatrix}2\\4\end{bmatrix}$, $\overrightarrow{b_1} = \begin{bmatrix}1\\-1\end{bmatrix}$, $\overrightarrow{b_2} = \begin{bmatrix}-1\\-1\end{bmatrix}$

6. Can the following matrix equation provide a unique solution for the scalar unknowns a *and* b? Justify your answer based on fundamental principle. [3 marks, CO01, BTL2]

$$\begin{bmatrix}1 & 1\\-1 & -1\end{bmatrix}\begin{bmatrix}a\\b\end{bmatrix} = \begin{bmatrix}12\\-12\end{bmatrix}$$

7. Consider an **operator / system matrix** $A = \begin{pmatrix}4 & 1\\-2 & 1\end{pmatrix}$. [14 marks, CO01, BTL5]

7.1 Determine the eigen values and eigen vectors of A

7.2 Use the results to obtain a set of orthonormal basis vectors for a 2×1 vector space.

7.3 Express the vector $\begin{bmatrix}10\\-20\end{bmatrix}$ as a linear combination of the orthonormal basis vectors obtained above.

***

Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	50	BTL 1	0
CO02	0	BTL 2	8
CO03	0	BTL 3	9
		BTL 4	14
		BTL 5	19
		BTL 6	0